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PRODUCT DOCUMENTATION

Modem



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Modem-M16

The Modem M16 is a two-way, half-duplex acoustic modem operating at 10bps that can function dependably in the presence of other modems configured on distinct channels. Boasting an impressively compact size, minimal power consumption, and affordable cost, the Modem M16 is a groundbreaking addition to the field. This innovation paves the way for successful underwater communication in complex environments, particularly where power resources for sensors are scarce. Buy Modem-M16 here!

Description

The Modem M16 is a two-way half-duplex 10bps acoustic modem with a robust and user configurable data link. The modem is delivered in four different versions, OEM, Flange, Standard, and Extended (RS422)

Mechanical

Information	All Modem M-16 Versions
Device length	47 mm (ex cable)
Device weight (air)	40 gram
Device weight (submerged)	14 gram
Depth rating	600 meter
Materials	POM (housing), PEEK (transducer)
Operating temperature	-10 to 70 °C
Device diameter	29.5 mm

Acoustics

Information	All Modem M-16 Versions
Range	1000 m
Transducer directivity	Omnidirectional
Transducer frequency	58.5 kHz

Information	All Modem M-16 Versions
Bits per package	16
Package duration	1.6 seconds
Max relative velocity	0.8 m/s

Relative movement and Doppler shift

The two modems should not move faster than **0.8 m/s relative to each other**. Above this relative velocity, the Doppler shift of the acoustic signal becomes significant and packets start to fail. See How fast can the modems move relative to each other? for an explanation and practical advice.

Installation guide

The Modem M16 Installation Guide (PDF) shows the pinout and the wiring diagrams for the UART and RS422 versions. For the Flange and OEM versions it also lists what is included in the box, and gives the mechanical drawings and the assembly steps.

Quickstart Guide

To start using the Modem-M16 you need

- a connection between the Modem and your computer
- a serial connection interface

To send and receive data between two Modem-M16's:

- both modems has to be in water (or less than 5 cm apart in air)
- both modems must be connected to a computer
- both modems need to be configured to the same channel.
- A sufficient power level must be used.

For an overview of the modems setting see Modem M16 Settings interface For implemetation of the modems settings see Modem M16 Protocol

---

By default the Modem M16 is delivered with the following settings:

- **Operational Mode:** Transparent serial mode
- **Communication Channel:** 1
- **Power level/Range Mode:** 4

How to connect to the Modem-M16

For wiring details see: Modem-M16 Wiring and the Modem M16 Installation Guide (PDF).

Connect using UART:

- Connect VIN to a 3.3V power source, and ground to ground/VOUT. The power source must deliver enough current when the modem transmits. Do not use the VCC pin of a USB-to-UART adapter, see Troubleshooting.
- Connect to your computer for instance using a UART to USB cable.

Connect using RS422:

- Connect VIN to a 12V (10-30V) power supply, and ground to ground/VOUT.
- Connect to your computer using an RS422 to USB serial converter.

Open up a user interface to interact with the modem

Open up a serial interface on your computer. This is typically Minicom on Linux and Putty on Windows.

It is also possible to use the designated GUI that can be installed on your computer. An installation guide is available here: Basic GUI for easy implementation

---

Configure the Modem-M16

When you are connected to a modem and have an open user interface, you will be able to read and write to configure the modem. The most basic configurations you need to know about are:

Operational mode There are two operational modes: - Transparent mode, that only processes and transmits two bytes of data - Diagnostic mode, that processes and transmits two bytes of data, and sends a detailed status report of the modem.

Changing channel For two modems to be able to communicate, they need to be configured on the same channel.

Range mode The modem have four different range modes or power levels, numbered 1 to 4, where 1 have the shortest range and 4 have the longest range. Shorter range uses less power during operation.

More information about the serial protocol and possible configurations of the M16, can be found here: M16-serial-protocol

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Start communicating between two modems

To be able to see what you send and receive on the modems you will have to open up one interface for each modem. Adjust the com/tty ports as necessary.

Begin sending data using either the GUI or the serial protocol; the modems will relay up to 2 bytes each ~1.6 seconds.

> **Note**
>
> To do configuration changes using the protocol, the letters needs to be sent with almost exactly 1 second between each letter. In any other circumstances the letters will be interpreted as a packege to be transmitted.

Make your own interface that suits your needs

Use the serial protocol M16-serial-protocol, and the Example Python Report Parser (described under Protocol) to create your own application that suites your needs.

Wiring interface

The tables below shows the pinning of the Modem-M16 interface. Depending on which type is used, the wires may have different colors.

The Modem M16 Installation Guide (PDF) has the same pinout and wiring diagrams, and the mechanical drawings for the Flange and OEM versions.

Interface	Colour (OEM)	Colour (Standard)	Colour (Extended)
VIN	Orange (3V - 4.2V)	Orange (3V - 4.2V)	Orange (10V - 30V)
GND	Orange-White	Orange-White	Orange-White
UART RX	Blue	Blue	NA
UART TX	Blue-White	Blue-White	NA
Shutdown (Active Low)	Green	Green	NA
RS422 RX+	NA	NA	Blue
RS422 RX-	NA	NA	Blue-White
RS422 TX+	NA	NA	Green
RS422 TX-	NA	NA	Green-White

Power consumption

Interface	Input voltage	Transmit (Tx)	Receive (Rx)
UART (OEM, Standard, Flange)	3.0–4.2 V DC	< 1.1 W	< 250 mW
RS422 (Extended)	10–30 V DC	< 2.0 W	< 400 mW

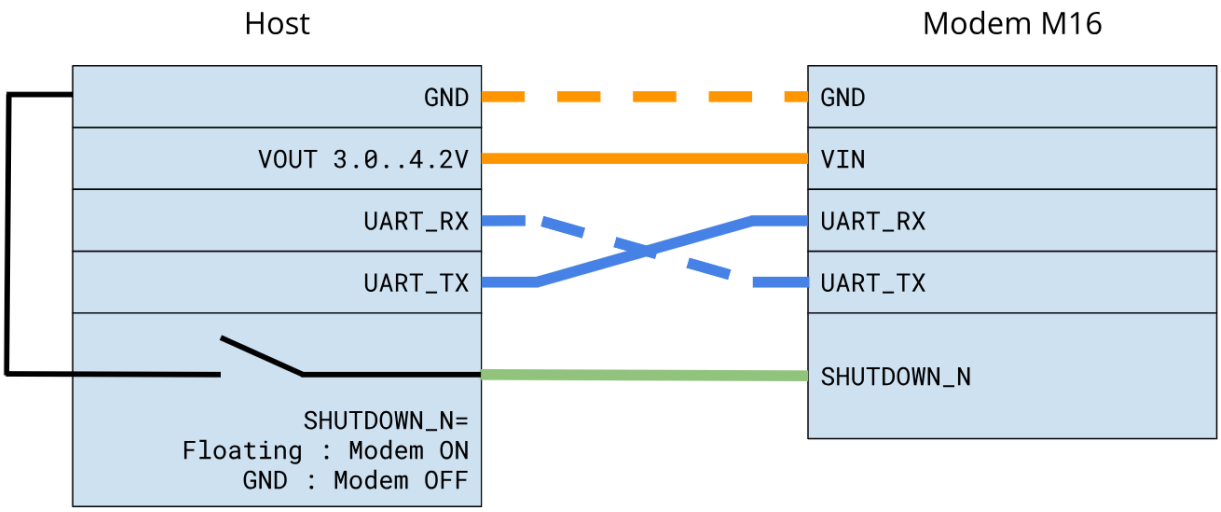
Transmit power depends on the selected power level, see Power Levels.

Note

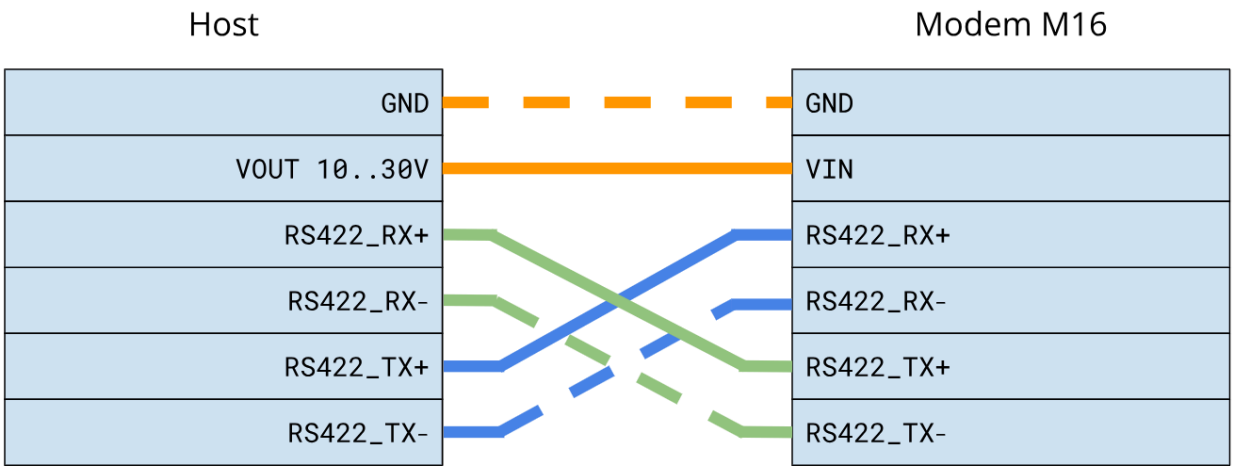
Shutdown line can be used to control the ON/OFF state of the Modem. If Shutdown is left floating, the Modem is ON. If Shutdown is short to GND, the modem is OFF. Shutdown line can be controlled by an external open-drain, open-collector or relay device. DO NOT apply voltage to Shutdown line.

The boot up time when powered on or toggling the shutdown line is approximately 12 ms.

Wiring OEM/Standard



Wiring Extended (RS422)



GUI

Basic GUI for easy implementation

For simple operation and tests there is a GUI available for download at: github for examples with Modem M16
When you follow the link, in the right menu, under Releases, press "Release 2025.06.12-1045

This executable provides a user-friendly interface to test and verify modem functionality. It demonstrates key features and allows for quick setup and communication testing between modems. Additionally it is a good starting point to get to know the functionality of the modem without Python. The app is a tkinter app that communicates with the modem through m16_driver.py.

GUI installation for windows:

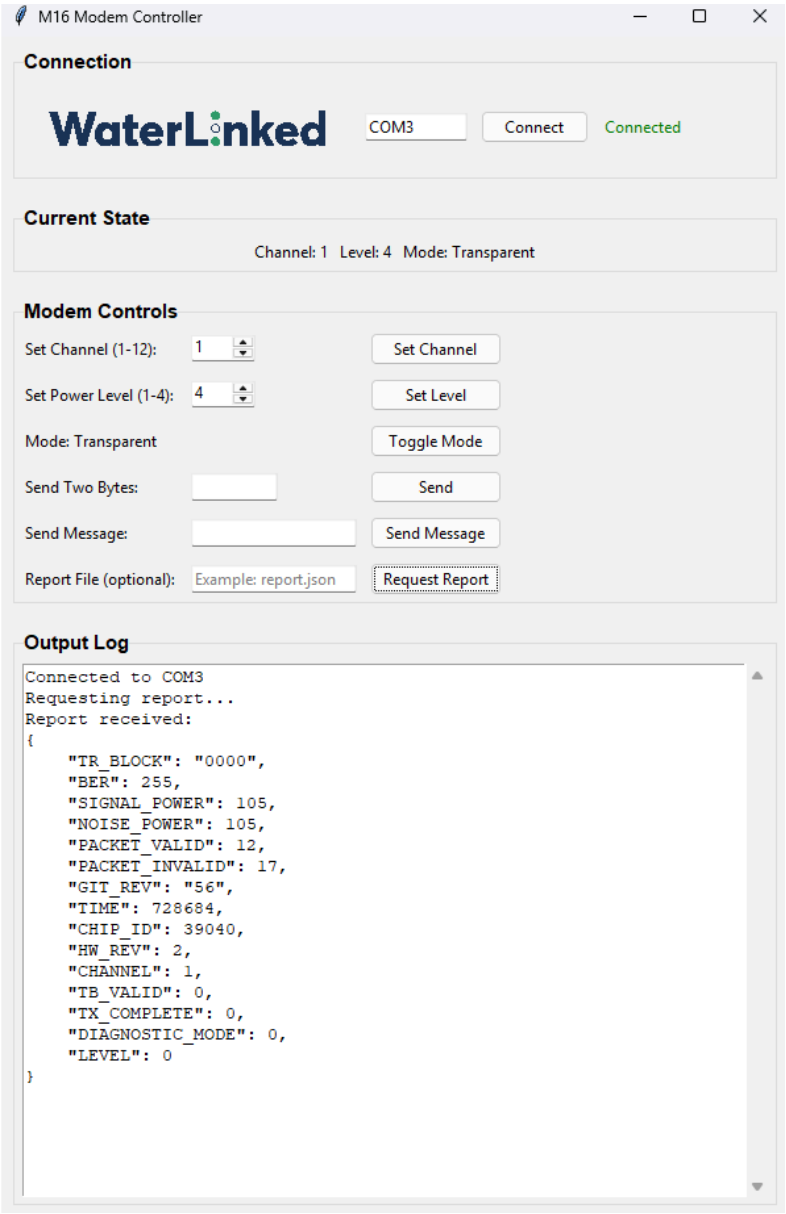
- Under "Assets", download gui.exe
- If window complains about the security you need to press "more info" and "allow"
- A powershell window will open up - just wait
- The GUI opens up and is ready for use

The default port is set to COM3, the user may need to change this to the appropriate port.

GUI installation for linux:

Follow the installation guide provided in README at github for examples with Modem M16

GUI layout example:



Integration

This document describes an overview of how the underwater modem can be operated. The modem can be controlled using a 3.3V UART or RS422 connection. It is offering two primary operational modes (transparent mode and diagnostic mode), four different power levels, user-selectable communication channels, and a possibility to set the modem in parrot mode for testing.

For details of the protocol used to implement the settings, see Modem-M16-Protocol

For the electrical and mechanical integration, see Wiring interface and the Modem M16 Installation Guide (PDF). The guide has the pinout, the wiring diagrams for the UART and RS422 versions, and the mechanical drawings and assembly steps for the Flange and OEM versions. Make sure the modem gets enough power, see Troubleshooting.

Interface overview

The standard Modem-M16 has a **3.3V TTL UART** interface and the extended version has a **RS422** interface. Both with the following configuration:

Settings	Value
Baud rate	9600
Data parity stop	8N1
Flow control	None

Through this serial link, you can:

- Set operational modes
- Change communication channels
- Request diagnostic reports
- Adjust power levels for transmission range/power consumption
- Set modem in parrot mode for testing

Settings Interface

You can configure the modem by issuing commands over the serial link. Five user-configurable parameters are available:

1. **Communication Channel**

2. **Operation Mode**
3. **Power Level**
4. **Report Request**
5. **Parrot mode**

Each configuration command requires sending specific characters with a 1-second pause between them, followed by a character that selects the new value. For details on how to send the commands, see the configuration protocol:Modem-M16-Protocol

---

Communication Channel

The modem can operate on **12 channels** (1 to 12). This allows for several modems to be used at the same time. Two modems must share the same channel to communicate. There are two ways to implement communication between several modems. You can have several sending modems that transmit on different channels, and one receiving modem that switch between the channels to listen for messages, or you can have several sending and receiving modems, where every two modems are sending and listening to each other on the same channel.

Operational Modes

The modem supports two main operational modes:

Transparent Serial Mode

- Processes and transmits only **2 bytes** of data per packet.
- Each transmission takes ~1.6 seconds (acoustic wave duration).
- Data beyond the first 2 bytes in a transmission is discarded.

Diagnostic Mode

- Sends detailed status reports, the same report that can be retrieved with the "request report" command, including:
  - Transport Block (TB)
  - Bit Error Rate (BER)
  - Signal & Noise Power
  - LDPC/CRC Validity
  - Firmware Revision
  - Time Since Power-Up
  - FPGA Chip ID, Hardware Revision
  - Communication Channel
  - TX-Complete Status
  - Power Level Setting
- Still limited to **2 bytes** of payload per transmission (discarding any additional data).
- Reports are triggered automatically when:

a. A new packet is received (TB Valid bit is set)

b. A transmission completes (TX Complete bit is set)

c. 4 seconds have elapsed since the last report

> **Note**
>
> You can also request a **diagnostic report** while in transparent mode. See the Request Report Command section.

Power Levels

The modem has four power levels where 1 is the weakest and 4 is the strongest level. If you need to save power and do not require the signal to travel a long distance, you might want to choose a lower power level. By default the modem is set to maximal power and range. Up to 1000m.

Power Level	Avg. TX Power (mW)	Typical Max Range (m)
1	390	300
2	555	600
3	790	900
4 (Default)	1105	1000

> **Note**
>
> Maximum range is highly dependent on environmental factors, including noise levels and line-of-sight conditions between modems.

Request Report

The diagnostic report provided by this command is the same as the report retrieved every fourth second in diagnostic mode.

In **both** diagnostic and transparent modes, you can request a report at any time.

When a command is sent the modem will immediately generate a diagnostic report.

For details of the structure of the diagnostic report and how it can be decoded see Diagnostic report

Parrot mode

A modem can be set to parrot mode, which might be useful for tests. When a modem is in parrot mode it always immediately returns the acoustic message it receives. This can typically be used in range tests with one modem set to parrot mode and one modem sending an acoustic message and wait for it to be returned.

> **Note**
>
> Do not forget to reset parrot mode after use.

Ensuring Successful Transmission & Optimization

Diagnostic Mode

A report is sent immediately after each transmission.
Monitor the **TX_COMPLETE** bit in the report to know when the modem is ready for the next 2-byte packet. This approach maximizes overall throughput.

Transparent Serial Mode

Send 2 bytes, then wait at least **1.6 seconds** (the transmission duration) before sending the next 2 bytes to avoid data loss.

Power-Up and Initialization

When powering on the modem or releasing the shutdown line from ground, it takes approximately **12 ms** for the modem to boot up and be ready to receive serial commands or data. Attempting to communicate with the modem during this 12 ms window may result in lost data.

Flash Memory Considerations

Changing the **modem mode**, **channel**, or **power level** initiates a **flash write operation**, which takes about **1 second**. During this time:

- The modem will **not respond** to commands.
- **Do not power off** the modem to avoid corrupting settings.

Once the flash write completes, the modem resumes normal operation with the updated configuration.

Firmware Release Notes

Below is a summary of the new features introduced in every firmware release:

FW release (ASCII/HEX)	Description
o/6f	Initial firmware release
V/56	Added power-level configuration for extended battery runtime and reduced range

Protocol

Modem-M16 Protocol

You can configure the modem by issuing commands over the serial link. Five user-configurable parameters are available:

1. **Communication Channel**
2. **Operation Mode**
3. **Report Request**
4. **Power Level**
5. **Parrot mode**

Each configuration command requires sending specific characters with a 1-second pause between them, followed by a character that selects the new value. Refer to the relevant section below.

> **Note**
>
> The timing of the commands need to be separated by exactly 1 second, if not it will interpret the input as a package to be sent. Doing this by hand is almost impossible, so it is advised to make a programmed user interface, or use the provided GUI (see WL GUI).

Communication Channel Command

The modem can operate on **12 channels** (1 to 12). Two modems must share the same channel to communicate.

Procedure:

1. Send `"c"` (0x63).
2. Wait **1 second**.
3. Send another `"c"` (0x63).
4. Within **2 seconds**, send **one character** (`"1"` – `"c"`) to set the desired channel.

Channel	Character	HEX-Value
1	"1"	0x31
2	"2"	0x32
3	"3"	0x33
4	"4"	0x34

Channel	Character	HEX-Value
5	"5"	0x35
6	"6"	0x36
7	"7"	0x37
8	"8"	0x38
9	"9"	0x39
10	"a"	0x61
11	"b"	0x62
12	"c"	0x63

Example (Channel 11): 1. Send `"c"`
2. Wait 1 second
3. Send `"c"` again
4. Within 2 seconds, send `"b"` (0x62)

Setting Operational Mode

To toggle between **diagnostic mode** and **transparent serial mode**:

1. Send `"m"` (0x6D).
2. Wait **1 second**.
3. Send `"m"` again.

The modem will switch modes (e.g., if you were in transparent mode, it will switch to diagnostic).

Setting Power Level

To adjust the modem's power level (1–4):

1. Send `"l"` (0x6C).
2. Wait **1 second**.
3. Send `"l"` again.
4. Within 2 seconds, send one character (`"1"` , `"2"` , `"3"` , or `"4"`).

Power Level	Character	HEX-Value	Avg. TX Power (mW)	Typical Max Range (m)
1	"1"	0x31	390	300
2	"2"	0x32	555	600
3	"3"	0x33	790	900
4 (Default)	"4"	0x34	1105	1000

> **Note**
>
> Maximum range is highly dependent on environmental factors, including noise levels and line-of-sight conditions between modems.

Example (Set Power Level 3): 1. Send `"1"` (0x6C).
2. Wait 1 second.
3. Send `"1"` (0x6C) again.
4. Within 2 seconds, send `"3"` (0x33).

Request Report Command

In **both** diagnostic and transparent modes, you can request a report at any time:

1. Send `"r"` (0x72).
2. Wait **1 second**.
3. Send `"r"` (0x72) again.

The modem will immediately generate a diagnostic report.

Set parrot mode

In parrot mode the modem immediately return the message it just got. This feature can be useful mainly for testing of the modem.

To set parrot mode:

1. Send `"P"` (0x70).
2. Wait **1 second**.
3. Send `"P"` (0x70) again.

To reset parrot mode:

1. Send `"p"` (0x50).
2. Wait **1 second**.
3. Send `"p"` (0x50) again.

Diagnostic report

This document describes the structure of the diagnostic report and how to decode it.

Packet Structure of the Diagnostic Report

When in diagnostic mode (or upon a report request in transparent mode), the modem sends an **18-byte** report. The byte layout is:

Byte (bit)	Field	Description
0 (0:7)	`START_OF_FRAME (SOF)`	"$" (0x24)
1:2 (0:15)	`TB`	Transport block containing 2 bytes of received data.
3 (0:7)	`BER`	Bit error rate
4 (0:7)	`SIGNAL_POWER`	Relative signal power (0–255)
5 (0:7)	`NOISE_POWER`	Relative noise power (0–255)
6:7 (0:15)	`PACKET_VALID`	Indicates packet integrity: LDPC decode + CRC successful
8 (0:7)	`PACKET_INVALID`	CRC check failed
9 (0:7)	`GIT_REV`	Firmware revision (Git commit hash, 8-bit)
10:12 (0:23)	`TIME_FROM_BOOT`	Time since power-up in 10 ms increments
13:14 (0:15)	`CHIP_ID`	FPGA chip ID
15 (0:1)	`HW_REV`	Hardware revision (2 bits)
15 (2:5)	`CHANNEL`	Active communication channel (1–12)
15 (6)	`TB_VALID`	Transport block valid bit; indicates new TB data
15 (7)	`TX_COMPLETE`	Indicates that a transmission has finished
16 (0)	`DIAGNOSTIC_MODE`	1 if in diagnostic mode; 0 otherwise
16 (1)	`RESERVED`	Reserved bit

Byte (bit)	Field	Description
16 (2:3)	`POWER_LEVEL`	Current power level (0 = L4, 1 = L3, 2 = L2, 3 = L1)
16 (4:7)	`RESERVED`	Reserved bits
17 (0:7)	`END_OF_FRAME (EOF)`	New line (“\n”)

Warning

Signal Power & Noise Power
- Both range from 0 to 255.
- When the modem is idle (not transmitting or receiving), **Signal Power** reflects the background noise floor, which should match **Noise Power**.
- **During** packet decoding, `SIGNAL_POWER` should be **greater** than `NOISE_POWER` for a successful decode.

Practical Tip
- Record the Noise Power while idle as your baseline (noise floor).
- Compare that with Signal Power during active communication.
- The bigger the gap, the better the link quality. (Could be closer to each other)

Example Python Report Parser

```
def decode_packet(packet: bytes) -> Optional[Dict[str, Any]]:
    """
    Decode a packet received from the modem.

    Parameters
    ----------
    packet : bytes
        The raw packet (18 bytes) starting with '$' (0x24) and ending with '\n' (0x0A).

    Returns
    -------
    Optional[Dict[str, Any]]
        A dictionary with decoded values if valid, or None otherwise.
    """
    if len(packet) != 18 or packet[0] != ord('$') or packet[-1] != ord('\n'):
        return None

    # The 16 bytes after the '$' and before the '\n'
    data = packet[1:17]
    decoded = struct.unpack("<HBBBHBBBBBHBB", data)

    decoded_dict = {
        "TR_BLOCK": decoded[0],
        "BER": decoded[1],
        "SIGNAL_POWER": decoded[2],
        "NOISE_POWER": decoded[3],
        "PACKET_VALID": decoded[4],
        "PACKET_INVALID": decoded[5],
        "GIT_REV": decoded[6].to_bytes(1, "little"),
        "TIME": (decoded[9] << 16) | (decoded[8] << 8) | decoded[7],
        "CHIP_ID": decoded[10],
        "HW_REV": decoded[11] & 0b00000011,
```

```
        "CHANNEL": (decoded[11] & 0b00111100) >> 2,
        "TB_VALID": (decoded[11] & 0b01000000) >> 6,
        "TX_COMPLETE": (decoded[11] & 0b10000000) >> 7,
        "DIAGNOSTIC_MODE": decoded[12] & 0b00000001,
        "LEVEL": (decoded[12] & 0b00001100) >> 2,
    }

    return decoded_dict
```

Troubleshooting: no response from the modem

Use this guide if the Modem M16 does not answer on the serial port, for example if you get no report, no reply to configuration commands, or nothing is received.

The most common cause is that the modem does not get enough power. A modem can answer when it is connected one way, and stop answering when it is moved to another board or adapter. This is often because the new board cannot deliver enough current.

Work through the steps in order. Each step removes one possible cause.

1. Power the modem from its own supply

Do not power the modem from the VCC or 3.3 V pin of a USB-to-UART adapter, a development board or a single-board computer. These pins are often not made for the current the modem needs when it transmits, and the voltage can drop or be at the wrong level.

Power the modem from a separate lab power supply or battery that can deliver the power below:

Version	VIN (orange ●)	Transmit (Tx)	Receive (Rx)	Recommended test supply
UART (OEM, Standard, Flange)	3.0–4.2 V DC	< 1.1 W	< 250 mW	3.3 V, at least 0.5 A
RS422 (Extended)	10–30 V DC	< 2.0 W	< 400 mW	12 V, able to deliver over 2.0 W

Source: the Modem M16 product page. At 3.3 V, 1.1 W is about 330 mA.

Warning

Do not connect 5 V to VIN on a UART version. 3.0–4.2 V is the allowed range.

If you use a lab power supply with a current limit, set the limit high enough. A current limit that is too low makes the voltage drop when the modem transmits.

2. Use a suitable converter for the data lines

Connect the modem to your computer with a converter that matches the interface:

- **UART versions:** a USB-to-UART converter with **3.3 V logic level**. If the converter has a 3.3 V / 5 V jumper, set it to 3.3 V. The UART is 3.3 V TTL, see Interface overview.
- **RS422 version (Extended):** a USB-to-RS422 converter.

Connect only the data lines and GND from the converter:

Modem M16 wire (UART)	Connect to
VIN, orange	+ of the separate supply
GND, orange-white	GND of the separate supply **and** GND of the converter
UART RX, blue	TX of the converter
UART TX, blue-white	RX of the converter
Shutdown, green	Leave floating (not connected)

The GND of the supply and the GND of the converter must be connected together. Without a common GND the data lines do not work.

For the RS422 wire colours, see Wiring interface. The Modem M16 Installation Guide (PDF) has wiring diagrams for both the UART and the RS422 versions.

Note

Leave the Shutdown line (green) floating. If it is connected to GND, the modem is off. Do not apply a voltage to it.

3. Test with the Water Linked M16 driver

Before you test with your own software, test with our Modem-M16-Driver on GitHub. It is known to work, so it tells you if the problem is in the hardware setup or in your own code.

- **Windows:** download `gui.exe` from the latest release, see GUI. Select the COM port of the converter and connect.
- **Linux or Python:** follow the installation steps in the README of the Modem-M16-Driver repository, and run `sending_examples.py`. Enter the serial port when asked (for example `/dev/ttyUSB0`).

Then request a report. The GUI has a button for this. In `sending_examples.py` the report is requested first and printed.

- **You get a report:** the modem, the power and the wiring work. If your own software still gets no answer, compare it with the driver. A common mistake is the timing: configuration commands, such as the report request, must be sent with almost exactly 1 second between the two characters, see Protocol.
- **You get no report:** go back to step 1 and 2. Measure the voltage on VIN while you request the report. It must stay within the range in the table above.

4. Test communication between two modems

When both modems give a report, test that they can talk to each other:

- Put both modems in water, for example in the same bucket.
- Set both modems to the same channel.
- Send two bytes from one modem and check that the other modem receives them.

See Quickstart for the default settings.

5. Contact support

If the modem still does not answer, submit a support ticket with:

- The serial number and version of the modem (UART or RS422).
- How the modem is powered: the supply or battery, its voltage and how much current it can deliver.
- The voltage you measured on VIN while you requested a report.
- The converter or board you use for the data lines, and its logic level.
- A photo or drawing of the wiring.
- What the M16 driver showed (a screenshot of the GUI or the output of `sending_examples.py`).

Modem M16 FAQ

Below are some frequently asked questions about the Modem M16. Click on a question in the table of contents to the right or simply scroll down to read the FAQs.

1. Can the M16 modem measure the range/distance between two modems?

No. The modems are unsynchronized and the M16 is not designed for time-of-flight ranging. Range estimates based on its timing will have large errors, for two reasons:

- **Transmit side:** The exact time of the acoustic transmission is not available on the interface. The TX_COMPLETE flag is raised within approximately 25 ms after the acoustic transmission, but the exact delay varies.
- **Receive side:** The modem decodes packets continuously and does not use a sync word. The point where a packet is registered as received is therefore not fixed relative to the acoustic arrival, and can vary by up to roughly half the duration of the acoustic transmission.

In water, these timing uncertainties correspond to path errors from tens of metres up to several hundred metres, which makes time-of-flight ranging impractical.

However, the Signal to noise ratio (SNR) can give a rough indication of whether the other modem is near or far: a higher SNR generally means the modems are closer to each other.

2. Do the modem retransmit data packets?

Yes, If the modem is set to parrot mode (see Parrot Mode), it will retransmit all packets it has received.

3. Does the modem need line of sight to communicate between them?

Yes, they do.

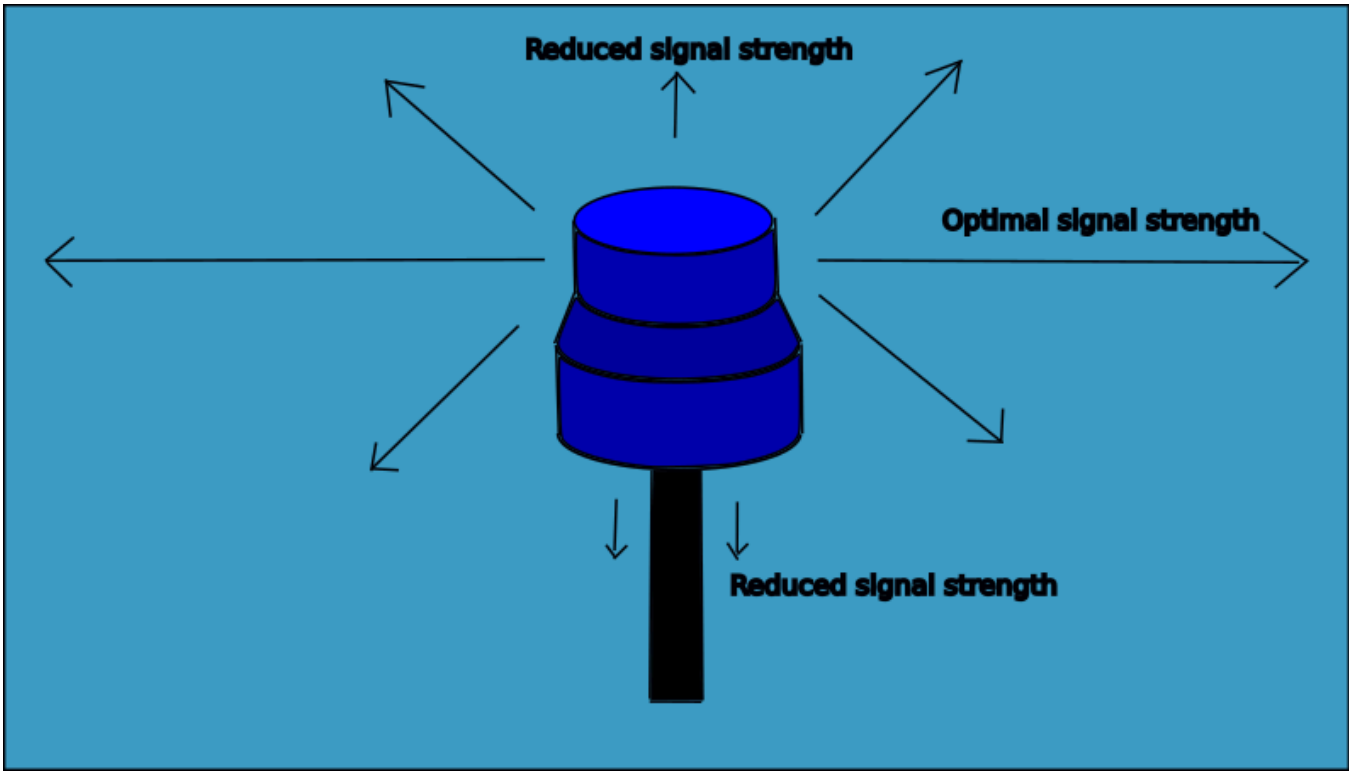
4. Can the modem be mounted below a shell on an ROV?

No, the transducer part of the modem must be in the water to be able to send and receive signals.

5. How should the modem be mounted to achiecve the best possible signal strength

The signal strength from the Modem-M16 is donut-shaped around the transducer.

Below is a simple scetch that illustrates the signal strength in all directions. For shorter distances the mounting is not relevant, as the signal is transmitted in all directions and also reflected at the surface, seabed or other constructions in the water. On long distances or to achieve optimal behaviour, the mounting position might slightly impact the signal range of the modem.



6. How fast can the modems move relative to each other?

The two modems should not move faster than **0.8 m/s relative to each other**. See Acoustics.

The limit comes from the Doppler effect. When two modems move towards or away from each other, the received signal shifts slightly in frequency. The M16 gets its long range on very little power by using a narrow frequency band and long packets, so it can only correct for a small shift. Up to about 0.8 m/s that works fine. Above that, packets start to fail and are dropped.

Two things help in practice:

- Only the speed directly towards or away from the other modem counts. Moving sideways around it has almost no effect.
- It is the relative speed between the two modems that matters, not speed over ground. Two vehicles traveling together at the same speed are not affected.

Situations like a winch lowering a modem quickly, or a boat moving up and down in waves, can be enough to go over the limit on their own. If you need to communicate while under way, slow down or hold position for the duration of the transmission, which is 1.6 seconds per packet.

7. The modem does not answer. What should I check?

The most common cause is that the modem does not get enough power. A USB-to-UART adapter or a development board often cannot deliver the current the modem needs when it transmits, even if the modem answers when it is connected another way.

Power the modem from its own supply or battery. For the UART versions, use 3.0–4.2 V (3.3 V is typical) that can deliver at least 0.5 A. For the RS422 version, use 10–30 V. Connect only TX, RX and GND from the adapter, and use a 3.3 V logic level for UART.

Then test with the Water Linked M16 driver and request a report. See Troubleshooting for the full step-by-step guide.

Modem-M64

The Modem-M64 is retired and no longer available for sale.

Description

The Modem M64 is a two-way half-duplex 64bps acoustic modem with a robust and user configurable data link.

The M64 sets a new market standard with its record-breaking physical size, low power requirement and cost. All this combined with a very useful technical specification, the M64 opens new possibilities for underwater communication in both existing and new areas.

LED Signals

- No green light: Power is off.
- Flashing green light (every 3 seconds): Modem is trying to pair with another modem.
- Fixed green light: Modem has sync with another modem. The LED will blink each time it receives a packet from the other modem (once a second).

Wiring interface

The tables below shows the pinning of the Modem-M64 interface. Depending on which cable type is used, the power wires may have different colors.

Interface	Color (Belden)	Color (4TP1P)
Positive (10-18V)	Orange	Red
Negative/Ground	Orange/White	Black
UART RX	Brown	Brown
UART TX	Brown/White	Brown/White

Note

Power must be applied to the power terminals before applying voltage to UART pins

Terminal Interface

The Modem-M64 has a 3.3 volt UART interface.

Settings	Value
Baud rate	115200
Data parity stop	8N1
Flow control	None

Channels

Modem M64 supports channels 1-7.

From (kHz)	To (kHz)	Channels using this frequency band
31.25	62.25	1
62.25	93.75	2 (16)
93.75	125.00	3 (16)
125.00	156.00	4 (17)
156.00	187.00	5 (17)
187.00	218.00	6 (18)
218.00	250.00	7 (18)

Please note that the Underwater GPS Explorer kit channels 16, 17, 18 overlaps and all channels overlaps with Underwater GPS G2.

Libraries and code examples

Example code and libraries that can be used to communicate with the modem on the terminal interface:

- Python

Dimensions



Protocol

Modem-M64 serial protocol

This document describes the Water Linked Underwater Modem Link Layer protocol.

Terminology

- Modem - Unit for transmitting/receiving data
- Packet - Unit of data transmitted together
- ACK - Acknowledgement. The command issued was successful.
- NAK - Negative acknowledgement. The command issued failed.

Version

This document describes protocol/software version 1,0,x (major,minor,patch)

The protocol versioning follows semantic versioning in that:

- MAJOR version increments represents incompatible API changes,
- MINOR version increments represents added functionality in a backwards-compatible manner
- PATCH version increments represents backwards-compatible bug fixes

Overview

The serial communication format is 115200 8-N-1 (no hardware flow control).

Packets sent to and received from the modem start with a `w` and end with LF or CR+LF. The packet format is:

Start byte	Direction	Command	Options (0 to many)	Checksum	End byte
`w`	`c` or `r`	`x`	`,[option]`	`*xx`	`\n` or `\r\n`

Direction is command (`c`) for commands issued to the modem and the modem replies with direction set to response (`r`). The commands can be sent as a string or entered one char at a time from a terminal.

The protocol can support Water Linked modems with different payload sizes and extended features. To support any Water Linked modem the connection procedure is to:

- Get protocol version. Verify that the major version number is 1.
- Get payload size. Use this size when queuing packets for transmission the modem.

For Water Linked Modem-M64 the payload size is 8 bytes.

In order for two modems to communicate they must be configured to use different roles (A/B) on the same channel. The modem with role A will always transmit. The modem with role B will listen until it detects a signal from a modem of role A. It will then start transmitting responses back to the modem with role A. The modem with role B will go back to listen mode if several consecutive packets from the modem with role A cannot be decoded (i.e. signal is lost).

Note

Payload where every byte is \0 is reserved. It is used to keep modems in sync if no data packet is queued by the user. This sync packet is filtered out by the receiver. Avoid this payload by compressing the data or otherwise ensuring atleast 1 bit is non-zero.

Checksum is optional when sending commands to the modem. The modem always returns a checksum. The checksum algorithm is CRC-8 and it is formatted as a hexadecimal number using 2 lower-case charaters (ex: `*c3`). See the checksum section of the manual for our DVL product (which uses the same CRC-8 algorithm) for further details and example code.

Commands

Commands in the table are shown **without** the checksum for readability.

Command	Description	Response	Description
`wcv`	Get protocol version	`wrv,` *[major],[minor], [patch]*	Protocol version. eg: `wrv,1,0,1`
`wcn`	Get supported payload size	`wrn,` *[size]*	Where size is supported payload size: eg: `wrn,8`
`wcc`	Get modem configuration	`wrc,` *[role],[channel]*	Where role=a or b and channel=1-7 eg: `wrc,a,3`
`wcs,` *[role], [channel]*	Set modem configuration - Where role=a or b and channel 1-7: eg: `wrs,b,3`	`wrs,a` or `wrs,n`	ACK or NAK
`wcl`	Get transmit queue length	`wrl,` *[q]*	Number of packets currently queued for transmission. eg: `wrl,107`
`wcf`	Flush transmit queue	`wrf,a` or `wrf,n`	ACK or NAK
`wcd`	Get diagnostic	`wrd,` *[link], [packet_count], [packet_loss_count], [bit_error_rate]*	link=y if connection with other modem, otherwise n. eg: `wrd,y,1234,17,3.5` *[1]*

Command	Description	Response	Description
`wcq,` *[size], [payload]*	Queue packet for transmission. Number of characters in payload need to be exactly same as size. Payload can be binary. eg: `wcq,8,HelloSea` *[2]*	`wrq,a` or `wrq,n`	ACK or NAK
		`wrp,` *[size],[payload]*	Got packet from other modem eg: `wrp,8,Welcome!`
		`wr?`	Malformed request: Response when packet cannot be understood
		`wr!`	Malformed request: Packet does not match the given checksum

[1] `packet_count` is the number of packets received. If a packet cannot be decoded the `packet_loss_count` increments regardless of the packet being queued is data or a sync packet from the other modem.

[2] Payload with just zeros (`\0`) is reserved for keeping the modems in sync if no packet is queued. It will be filtered out by the receiving modem.

Examples

Here is an example of setting up two modems and sending packets between them.

On top side modem (using role A):

Command	Response	Description
`wcv`	`wrv,1,0,1*44`	Get protocol version
`wcn`	`wrn,8*ba`	Get payload size
`wcs,a,3`	`wrs,a*01`	Set role A and channel 3
`wcq,8,HelloSea`	`wrq,a*d7`	Send packet
		Wait packet from other modem
	`wrp,8,HelloTop*bb`	Got response

On AUV/ROV modem (using role B):

Command	Response	Description
`wcv`	`wrv,1,0,1*44`	Get protocol version
`wcn`	`wrn,8*ba`	Get payload size
`wcs,b,3`	`wrs,a*01`	Set role B and channel 3
		Wait packet from other modem
	`wrp,8,HelloSea*58`	Got packet
`wcq,8,HelloTop`	`wrq,a*d7`	Send response back